Unsupervised Patch-Based Cloud Classification of SEVIRI Imagery

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1 Abstract

We develop an unsupervised patch-based, adaptive binary classifier of SEVIRI (satellite) imagery which is motivated by its importance for downstream remote-sensing applications and for pre-processing meteorological applications. For each region (300 km $\times$ 300 km), pixel value histograms in one visible (Ch2) and three infrared channels (Ch7, Ch9, and Ch10) are modeled using the Bayesian Gaussian Mixtures Model. Individual decision thresholds are chosen for each channel through consideration of local minima in the mixture probability density, and the overall decision mask is the logical intersection of each channels' threshold. On a hand-labeled oceanic patch containing optically thick clouds, the method achieves 91.1% accuracy, 93.7% precision, and 93.1% recall, whereas on a thin-cirrus ocean patch recall drops to 53.4% while precision remains 90.0%. The approach is therefore robust with a low number of false positives, and performs well for obvious cloud regimes but requires more specialised spectral analysis for optically thin and low-contrast surfaces.

2 Introduction

For this mini-project, we aim to classify pixels over regions of Earth as cloud/not cloud (binary classification) by developing an unsupervised Bayesian Gaussian Mixtures model applied to a few selected infrared and visible channels where decision boundaries are adaptively chosen based on the patch of Earth we are classifying. The limitations and performance of this approach are also analyzed across multiple surface types.

The binary classification of cloud pixels have a wide-range of use cases and is mainly used as a pre-processing step for other more "practical" use cases of satellite imageries. Examples of this could be its use in remote-sensing workflows where you would want to perform analysis on the Earth's surface, such as to monitor vegetation health (a common metric is the Normalized Difference Vegetation Index - NDVI) and so corrections or different methods entirely would have to be used in cloudy conditions (i.e. Qin et al. 2024).

EUMETSAT, the European Organisation for Meteorological Satellites, have a satellite called the Meteostat Second Generation (MSG) which is in a geostationary orbit at 0° East (in standard latitude/longitude coordinates). The Spinning Enhanced Visible and Infrared Imager (SEVIRI), which is the radiometer aboard the MSG, can capture images on 12 different wavelength channels spanning the visible and infrared spectrum (seen in Table 1).

Table 1: SEVIRI Channels (EUMETSAT 2020)
The channels probe/cover the following spectra: VIS (visible), NIR (near-infrared), IR (infrared), WV (water vapour), O3 (ozone), CO2 (carbon dioxide)

Channel	**Wavelength**
Ch1 VIS	0.64 μm
Ch2 VIS	0.81 μm
Ch3 NIR	1.63 μm
Ch4 IR	3.92 μm
Ch5 WV	6.31 μm
Ch6 WV	7.36 μm
Ch7 IR	8.71 μm
Ch8 O3	9.67 μm
Ch9 IR	10.79 μm
Ch10 IR	11.94 μm
Ch11 CO2	13.35 μm

Clouds generally are of lower temperature and higher reflectance[1] than neighbouring surfaces — this is the main assumption of our experiments. However, this is just a rule of thumb as exceptions occur for clouds poorly contrasting with neighbouring surfaces such as thin cirrus clouds (which are typically warmer) or polar ice regions (where the surface can match cloud temperatures), as discussed in Amato et al. 2008. As such, it is important we use an aggregation of spectral channels, which will help the performance of our statistical classifier. In this report we will restrict ourselves to a handful of channels since the scaling of the number of features introduces challenges associated to the curse of dimensionality and a more complicated task ahead. Papers such as Amato et al. 2008 address this problem more comprehensively with methods such as principal component analysis, which also allows them to make full use of all spectral channels available.

This report is structured as follows. In section 3, we provide the methodology used for determining whether a given pixel is a "cloud" pixel or a "not-cloud" pixel. Then in section 4.1 we visually show how our method works, and then show numerical results in section 4.2 using our validation method as described in section 3.5. We then conclude the report with limitations and perhaps any future work that could be done.

3 Methods

3.1 Channel selection

To tackle a simpler problem, we would like to restrict ourselves to a few channels of interest. Generally speaking, the radiance received by Seviri is a weighted average of the radiances of the surface[2] and of the radiances of the atmosphere, where the average is weighted by the atmospheric absorption. The absorption values have a strong dependency on wavelength, and so it is important to consider the absorption spectrum of H_2O as seen in Figure 1.

[1] Reflectance is defined as the ratio of reflected to incident electromagnetic power.

[2] See equations 3 & 4 of the ap03 lab script `https://www-teaching.physics.ox.ac.uk/practical_course/scripts/srv/local/rscripts/trunk/Atmos/AP03/AP03.pdf`

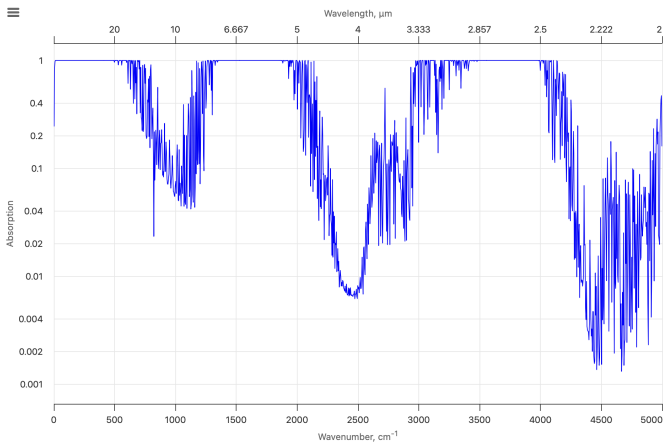


Figure 1: Plot of the absorption spectrum of H_2O from Earth Observation Data Group 2026
Note the logarithmic y-axis

SEVIRI channels were chosen such that the absorption strength of water (cloud) molecules is high enough so that the cloud could be probed, whilst not too high such that atmospheric vapour reduces contrast with adjacent cloud and surface pixels. The channels chosen should thus give a high enough contrast for signal received from neighbouring cloud and surface pixels, as our methodology relies on this. A brief inspection of Figure 1 and some exploratory data analysis led us to select Ch2 (0.81 µm), Ch7 (8.71 µm), Ch9 (10.79 µm) and Ch10 (11.94 µm) — which all have absorptions on the order of ~ 0.1. And Ch1 (0.64 µm) was used for visuals, since to avoid bias, we can't hand-label pixels using the same channels we'll be training on.

Ch2 supplies our reflectance contrast information (i.e. high radiance implies a high likelihood of it being a cloud). Ch7, Ch9 and Ch10 were selected as general purpose cloud detection channels given that they lie in the "middle" range of absorption strength compared to all other channels — multiple channels were chosen in the hopes to add some robustness to the classification. For Ch2, we use its radiance, B, as our feature[3], but for Ch7, Ch9, and Ch10, we use their "brightness temperature" which is simply a result of inverting the Planck function for temperature, T, as a function of radiance as shown in equation (1) where all other constants have their canonical meaningsw.

$$B(\lambda, T) = \frac{2hc^2}{\lambda^5(\exp(hc/\lambda k_B T) - 1)} \Rightarrow T(\lambda, B) = \frac{hc}{\lambda k_B \ln(1 + 2hc^2/\lambda^5 B)} \tag{1}$$

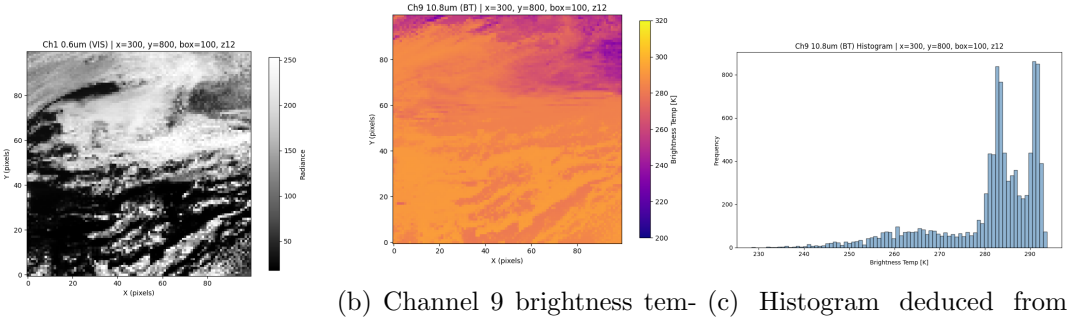
We note that all manual validation is done by viewing Channel 1 (which is pure visible channel), and as such, we do not use it as an input to our classifier to remove any implicit biases that may be present.

3.2 Pixel Value Histograms

Through some exploratory data visualisation, one can see that distinct features (in the visible channel 1) on a patch of Earth usually leads to distinct peaks in the histogram of

[3]In Machine Learning literature, a "feature" is one data type used to train a model - i.e. someone's age for predicting their health.

all pixel values on that patch of Earth. An example of this can be seen in figure 2c for channel 9 brightness temperature.



(a) Channel 1 radiance

(b) Channel 9 brightness temperature

(c) Histogram deduced from figure 2b

Figure 2: Example patch of Earth (100px by 100px) showing how the visually distinct clouds and sea provides distinct peaks on the corresponding histogram (for channel 9)

From looking at Figure 2, we can see empirically, that each class (i.e. ocean or thick cloud) show broad unimodal peaks which we model as gaussians.
It is for this reason, we attempt to decompose the histograms into their component gaussians to essentially identify each distinct feature of this patch of Earth. Once we can identify each individual gaussian making up the total histogram, we can then use the (simple) physical principles that the surface will warmer and less bright than the clouds. So for example, looking at figure 2c, the rightmost "gaussian" with its peak at 292K is almost surely the ocean while the gaussian centred at ~ 285K corresponds to clouds.

3.3 Bayesian Gaussian Mixture Model

To tackle the problem of modeling the histogram as an ensemble of gaussians where the number of gaussians, their means, and variances are unknowns, we use an unsupervised machine learning model known as the Bayesian Gaussian Mixture Model (Bayesian GMM). What follows is a brief explanation of how it works with more details in chapters 9.2 and 10.2 of Bishop 2006.

3.3.1 Standard GMM

We imagine building up the histogram from our model, so that if our model is correct, then sampling from it should recreate something with the same statistical properties as the histogram. We first assert there are K gaussians where we select the ith gaussian with probability π_i. This gaussian will have mean, μ_i, and variance σ_i from which we then generate our data point: $x \sim \mathcal{N}(\mu_i, \sigma_i^2)$. Thus the probability of observing a data point x, given our parameters $\theta = \{\pi_k, \mu_k, \sigma_k\}_{k=1}^K$ is given by equation (2)

$$p(x \mid \theta) = \sum_{k=1}^{K} \phi_k \mathcal{N}(x \mid \mu_k, \sigma_k) \tag{2}$$

And so we want $\hat{\theta} := \arg\max_\theta \prod_{n=1}^N p(x_n \mid \theta)$[4] where N is the number of pixels[5] (data points) in our sample — i.e. a maximum likelihood estimation for our parameters.

[4] $\arg\max_\theta f(\theta)$ returns the θ that maximises f.
[5] One pixel represents a 3 km $\times$ 3 km region in the SEVIRI dataset.

3.3.2 Bayesian GMM

In standard GMM, we seek the probability of getting our dataset given our parameters: $p(X \mid \theta) \equiv p(\text{data} \mid \text{parameters})$ and try to find the optimum θ. In Bayesian GMM, we want to find the probability of having a set of parameter values given our data: $p(\theta \mid X)$. In this way, what we get is a probability distribution over all models. We then obtain a prediction for a single data point x in equation (3) using the total law of probability[6].

$$p(x \mid X) = \int p(x \mid \theta)\, p(\theta \mid X)\, \mathrm{d}\theta \tag{3}$$

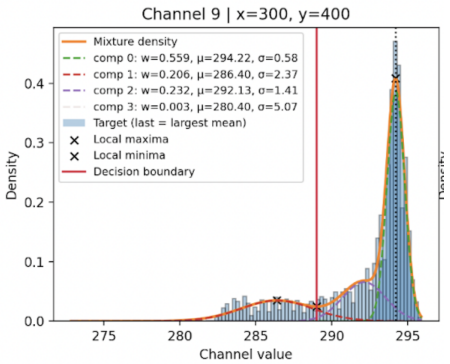
For our paper, we use the Python implementation from scikit-learn (scikit-learn 2025) to find the optimal set of parameters.

3.4 Decision Boundary

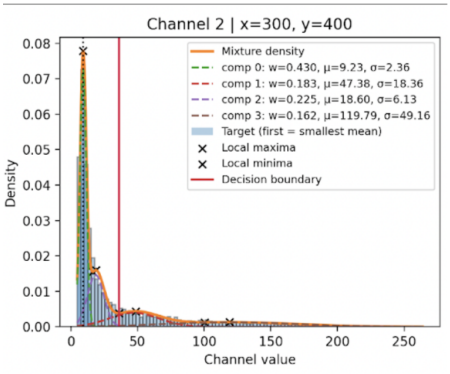
In this report, we take the upper limit to the number of gaussians to be 4 to avoid over-computing.
Now in deciding how to construct decision boundaries[7], one thing to be mindful of is the fact that 2 gaussians may overlap each other in such a manner that they may not actually represent 2 distinct features. Thus we consider the combined mixture probability density function (pdf), which is just a superposition of all gaussians, and take the threshold[8] value to be at where the first local minimum is found. This can be seen pictorally in figure 3a.

Note that for channels 7,9,10, we define the "first" local minimum by counting from the right (i.e. from the highest temperature) whereas for channel 2, we look from the left (i.e. from the lowest radiance). This is to conform to our physical principles that clouds are lower temperature and higher reflectance than the surface.

Figure 3: The sum of the 4 gaussians give the mixture density. Each gaussian (along with μ, σ have a weight w which is its probability π.



(a) "Channel value" is Brightness Temperature. Can also note how the Bayesian GMM can only plausibly find 3 gaussians as we can see the weight for comp 3, w, is very small (0.003).



(b) "Channel value" is Radiance. Notice how the first local minimum is skipped because it cannot be meaningfully resolved.

[6] We use $p(x \mid \theta, X) = p(x \mid \theta)$ since once given parameters, x and X are independent.
[7] The conditions at which we decide whether to classify as a cloud or not cloud.
[8] The threshold values for each channel are like mini decision boundaries, which will be used to create our final decision boundary, seen in section 3.4.1.

Sometimes we get a local minimum which is difficult to resolve like the one at ~ 18.6 in figure 3b. This essentially means that the minimum is close to the local maximum of the next gaussian, and so we can devise a criteria to ignore that minima if it is within 0.5 standard deviations of the local maximum.

3.4.1 Combining Channels

Now we have thresholds for each channel which allows us to create a mask[9], $\mathcal{M}_i$, for channel i. The final cloud mask $\mathcal{M}$ (which represents our decision boundary) is chosen to be the logical intersection[10] of cloud masks for our chosen channels. This reduces false positives by adding robustness, but this comes at a cost of reduced sensitivity to optically thin clouds (which can only be picked up in certain wavelengths, for example). This was chosen over other methods (such as majority voting) due to its simplicity and lack of false positives compared to other crude methods that we tested.

3.5 Validation

Since our dataset is unlabeled (i.e. we have no ground truth values), it is tricky to verify the correctness of our classifier. For the most part, we look at the mask the algorithm generates and compare to the image in visual channel 1 via a brief visual inspection. But to add some level of exactness, we hand-labelled[11] 2 different patches of Earth and report the performance of our algorithm in these patches in section 4.2.

4 Results and Discussion

4.1 Method Illustration

The region shown in Figure 4 is actually over land and is a 100 px by 100 px area which corresponds to a 300 km $\times$ 300 km area. Here we will briefly discuss why the algorithm seems to have worked fairly well and any shortcomings it may have.

We can see in the histograms the thresholds (the red vertical lines) that have been selected for each channel. For channels 7 and 9, the thresholds identify any pixel with a brightness temperature above 305 K as not part of a cloud. For channel 10, any pixel above 291 K is not a cloud. For channel 2, any pixel radiance below 105 $\mathrm{mW\,m^{-2}\,sr^{-1}\,cm^{-1}}$ is not a cloud.

We can see that the channel 7 and 9 thresholds are a bit too high, given that they allow anything below roughly 35°C to be classified as a cloud. Since our mask is the intersection of all channels, we are thankfully saved by channels 10 and channel 2 giving reasonable thresholds. Different channels have different transmittances and weightings, and so clouds of a given optical depth in certain regions may be more visible in one channel than in others. As such, having the intersection rule for our overall mask works well as a conservative approach which reduces the number of false positives.

[9] A mask assigns a value of 0 (no cloud) or 1 (cloud) to a pixel depending on where the pixel value lies in relation to the decision boundary.

[10] $\mathcal{M} = \bigcap_{i \in \{2,7,9,10\}} \mathcal{M}_i$

[11] Meaning we manually labelled each pixel as a cloud vs not cloud, still by inspecting visual channel 1

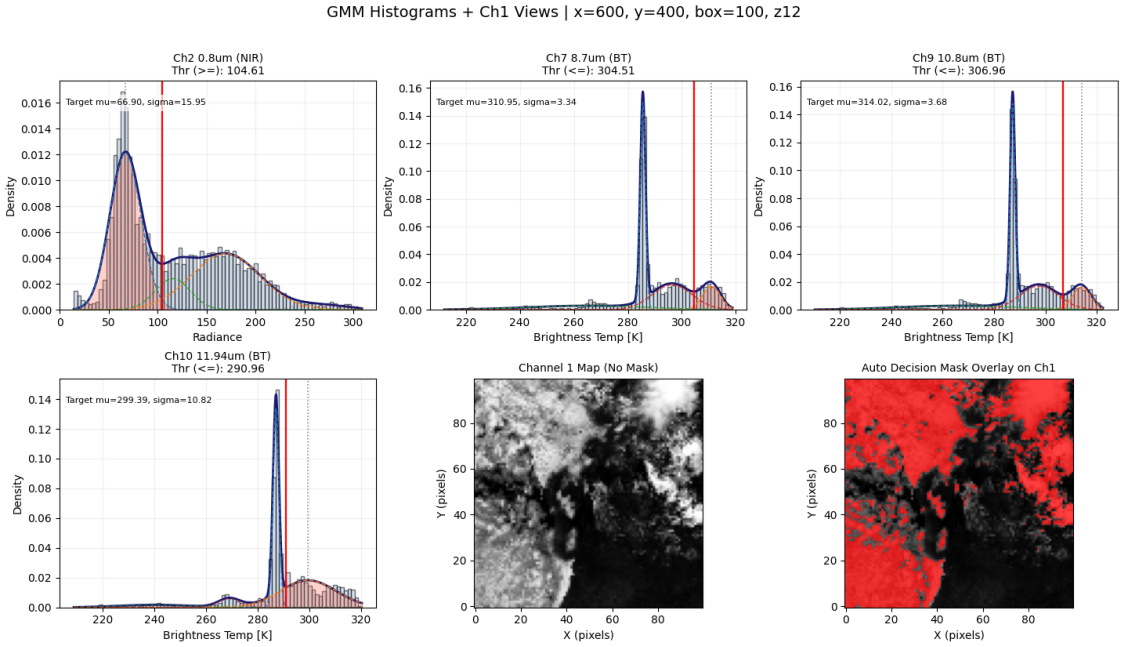
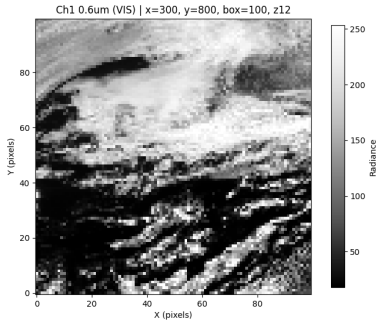


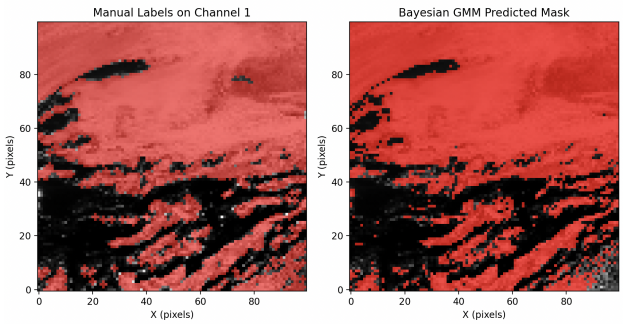
Figure 4: Thresholds for the Bayesian GMM is visualised in the first 4 plots for our channels and the visual channel is shown, with the mask coloured on as red indicating our model's cloud classification.
Patch centre: latitude $= -7.357556°$, longitude $= 15.402386°$. This is over land in Angola, just North-East of Luanda.

4.2 Numerical Validation

4.2.1 Prediction on cumulonimbus clouds



(a) Channel 1 visual for given region



(b) Left is the hand-labelled clouds (our "ground truth"), right is what the algorithm computed as our mask.

Figure 5: Patch centre: latitude $= 41.860609°$, longitude $= -25.3872536°$. This region corresponds to a patch in the North Atlantic ocean.

We can see through Figure 5 and Table 2, that our algorithm does quite well for "obvious" clouds i.e. cumulonimbus clouds. This is a given, since our algorithm benefits from distinct gaussians and it is clear from Figure 5a that the overall histogram would be a distinct bimodal distribution which makes it easy for the Bayesian GMM to identify

Table 2: Confusion matrix for Figure 5b — shows number of classifications/misclassifications for pixels. "True" here refers to the hand-labelled classification which is defined as our current ground truth

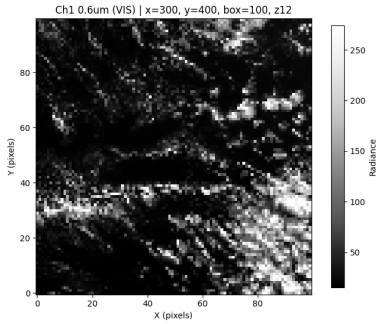
	Predicted No Cloud	**Predicted Cloud**
True No Cloud	2803	423
True Cloud	465	6309

clouds.

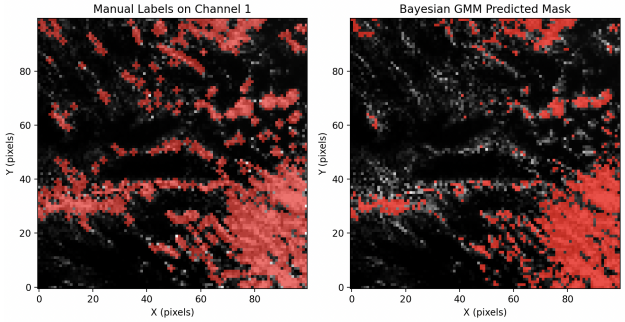
From Table 2 we can see that the number of false positives is low (false positives are the number where we predict a cloud, but the ground truth is that there's no cloud), which is a consequence of our algorithm using logical intersections. We also have a high accuracy (see Table 4) due to this being a type of region where our algorithm will perform best, as explained earlier.

However, one thing to note is that our hand-labelling is not completely 100% accurate (since it is done by eye, and some thin cirrus clouds may elude the hand-labelling) and as such, the numerical values should not be treated exactly, but heuristically.

4.2.2 Prediction on thin cirrus clouds



(a) Channel 1 visual for given region



(b) Left is the hand-labelled clouds (our "ground truth"), right is what the algorithm computed as our mask.

Figure 6: Patch centre: latitude = $-7.372959°$, longitude = $-17.786758°$. This region corresponds to a patch of ocean in the South Atlantic ocean.

Table 3: Confusion matrix for Figure 6b

	Predicted No Cloud	**Predicted Cloud**
True No Cloud	7034	165
True Cloud	1306	1495

Here, we see a major weakness of our algorithm in detecting these thin cirrus clouds. As we can see in Table 3, there is a significant proportion of clouds that were classified as

not clouds. But we can also note that the number of false positives is still low, which is again, a consequence of using the logical intersection for our overall mask.

Thin cirrus clouds are not as tall and not as dense as thick cumulonimbus clouds therefore they transmit a lot of the surface radiance hence temperature. Thus at least for the channels we have, which are not sensitive to small contrasts between clouds and neighbouring regions in an image, we simply cannot distinguish thin cirrus clouds with enough certainty from the ocean.

Generally, the algorithm is just not able to properly classify thin cirrus clouds resulting in many false negatives (i.e. in Figure 4 too). In order to identify thin cirrus clouds, one would have to analyse the wavelength specific to them rather than considering different, general purpose channels.

4.2.3 Performance Metrics

Table 4: Classification performance for the two manually labelled 100×100 patches, figures 5 and 6. Accuracy is the fraction of correctly classified pixels overall. Precision is the fraction of pixels classified as cloud that are truly cloud (i.e. high precision corresponds to low false-positive rate). Recall is the fraction of true cloud pixels correctly identified (i.e. high recall corresponds to low false-negatives).

Region	Accuracy (%)	Precision (%)	Recall (%)
Figure 5	91.1	93.7	93.1
Figure 6	85.3	90.0	53.4

From this, we can see the claim that our algorithm is robust (i.e. low false positives) due to the logical intersection mask is well-founded as seen from the high precisions and visual inspection (we rarely see the surface being classified as a cloud).

The low recall in figure 6 shows how our algorithm is not really equipped to dealing with all types of clouds (thin cirrus in particular) as it misses a lot of clouds and marks them incorrectly as not cloud.

4.3 Limitations

Our algorithm has a plethora of other limitations which are not shown in the results (due to space) but we will discuss some briefly here.

- Our algorithm assumes a homogeneous surface temperature and radiance. As such, our algorithm would fail in regions where there is land and sea, or in regions where the surface vastly changes (i.e. when grasslands meet a desert, the radiance and temperature would wildly vary). This would break the assumption that only one gaussian corresponds to the surface and the rest represent clouds. One potential fix is to assume clouds have a minimum radiance and a maximum temperature.
- Our algorithm was only used for imagea taken in day-time. For images taken at night, the radiance channel would essentially be useless and so a better option would be to use night-time channels like channel 4 for night (which acts as channel 2 would act during the day).

- There are regions where clouds do not have to have different radiances or temperatures from their adjacent surfaces in the images — namely polar,icy regions. This is the reason why the thin cirrus clouds were not detected (since the radiance/temperature received from thin cirruses is very similar to that of the adjacent surface).

5 Conclusion

In this report, we implemented the Bayesian Gaussian Mixtures Model to separate and classify pixel-value histograms over patches of Earth, and added decision thresholds over multiple channels to generate an overall mask to binary classify cloud pixels. Strong performance was achieved in optically thick clouds next to homogeneous surfaces (see section 4.2.1), indicating that the approach is able to generate reliable cloud masks.

A failure mode that was explored, was the models inability to identify thin cirrus clouds (see section 4.2.2), where the surface signal is transmitted more and the histogram separation becomes more ambiguous. Additional limitations include non-homogeneous surface emissivity and temperature (e.g. along coastlines), polar regions which have low contrast with the clouds, and the reliance on a visible channel during daytime. Future work could incorporate channels more sensitive to optically thin clouds, as well as channels that could be used during night-time (e.g. Ch4) to increase the model's accuracy and recall.

References

Amato, U. et al. (2008). "Statistical cloud detection from SEVIRI multispectral images". In: *Remote Sensing of Environment* 112.3, pp. 750–766. ISSN: 0034-4257. DOI: https://doi.org/10.1016/j.rse.2007.06.004. URL: https://www.sciencedirect.com/science/article/pii/S0034425707002313.

Bishop, C.M. (2006). *Pattern recognition and machine learning*. Vol. 4. Springer New York. URL: http://scholar.google.com/scholar.bib?q=info:jYxggZ6Ag1YJ:scholar.google.com/&output=citation&hl=en&as_sdt=0,5&as_vis=1&ct=citation&cd=0.

Earth Observation Data Group Department of Physics, University of Oxford (2026). *Zenith Absorption*. https://eodg.atm.ox.ac.uk/ATLAS/zenith-absorption. Accessed: 2026-02-22.

EUMETSAT (2020). *SEVIRI Instrument User Guide*. Tech. rep. European Organisation for the Exploitation of Meteorological Satellites. URL: https://user.eumetsat.int/resources/user-guides/msg-high-rate-seviri-level-1-5-data-guide.

Qin, Peng et al. (2024). "Reconstructing NDVI time series in cloud-prone regions: A fusion-and-fit approach with deep learning residual constraint". In: *ISPRS Journal of Photogrammetry and Remote Sensing* 218, pp. 170–186. ISSN: 0924-2716. DOI: https://doi.org/10.1016/j.isprsjprs.2024.09.010. URL: https://www.sciencedirect.com/science/article/pii/S0924271624003484.

scikit-learn (2025). *sklearn.mixture.BayesianGaussianMixture — scikit-learn 1.8.0 documentation*. Accessed: 2026-02-22. URL: https://scikit-learn.org/stable/modules/generated/sklearn.mixture.BayesianGaussianMixture.html.